

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**For example:**

| Input | Result |
|-------|--------|
| 9     | 12     |

**Answer:** (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  void function(int n)
4  {
5      int i = 1;
6      int s = 1;
7      int count = 3;
8
9      while (s <= n)
10     {
11         i++;
12         s += i;
13         count += 3;
14     }
15
16     printf("%d", count);
17 }
18
19 int main()
20 {
21     int n;
22     scanf("%d", &n);
23
24     function(n);
25
26     return 0;
27 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 9     | 12       | 12  | ✓ |
| ✓ | 4     | 9        | 9   | ✓ |

Passed all tests! ✓